Exact certification of a paired Jacobsthal bound and the dimension-2 sieve barrier: a reproducible note on a tightened Goldbach statement

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Draft v0.2 — 10 November 2026 — **no proof of Goldbach's conjecture is claimed**

Abstract

Ziller and Morack reduce a tightened form of Goldbach's conjecture to a bound $h_2(k) < p_k^2 - p_k$ on a paired Jacobsthal function. We (i) re-derive the reduction elementarily, (ii) certify the source's values $h_2(k)$ for $k = 3, \dots, 11$ (including $h_2(10) = 450$ and $h_2(11) = 570$) by an exhaustive solver-free search whose infeasibility proofs need up to 1.4×10^{10} nodes, (iii) check the tightened statement for all even $N \in [12, 2 \cdot 10^7]$, (iv) tabulate exact fixed-window minima M_k, and (v) explain, from source-reported sifting limits, why a standard dimension-2 sieve cannot reach the required range. The bound itself remains *unproved* for all k; rows $k \geq 12$ are transcribed. Novelty of every item is **not** established; see Section 6.

1 Setting

Let p_k be the k-th prime. For $k \geq 3$, $h_2(k)$ is the least m such that for every pair (a, b) with $2 \mid (b - a)$ some $q \in \{1, \dots, m\}$ makes both $a + q$ and $b + q$ coprime to $p_1 \cdots p_k$ [1, 2]. Equivalently, $h_2(k) = 6\omega_2(k) + 6$, where $\omega_2(k)$ is the largest L such that one can choose *two* nonzero residues modulo each prime $p \in \{5, \dots, p_k\}$ (one class mod 2, by the source's normalisation) covering all positions $1, \dots, L$ [3]. We use the source's equivalence as stated and verify it only through direct witness re-checks (Section 3).

Conjecture 1 (H; source Conjecture 6)**.** *$h_2(k) < p_k^2 - p_k$ for all $k \geq 3$.*

Conjecture 2 (Tightened Goldbach; source Conjecture 4)**.** *For even $N \geq 12$ with $p_k^2 + p_k \leq N < p_{k+1}^2 + p_{k+1}$ there are primes q_1, q_2, $p_k < q_1 < p_k^2$, $q_1 + q_2 = N$.*

2 The reduction

Proposition 1 (source Prop. 3.2, re-derived)**.** *For $k \geq 3$, if $h_2(k) \leq p_k^2 - p_k - 1$ then Conjecture 2 holds for every even N with $p_k^2 + p_k \leq N < p_{k+1}^2 + p_{k+1}$.*

Proof. Put $p = p_k$ and take $(a, b) = (p, p - N)$; $b - a = -N$ is even. There is $q \leq p^2 - p - 1$ with $p + q$ and $p - N + q$ both coprime to $p_1 \cdots p_k$. Let $q_1 = p + q$, $q_2 = N - q_1$. Then $p < q_1 < p^2$ and q_1 has no prime factor $\leq p$, so it is prime ($q_1 < p^2$). Next $q_2 \geq p^2 + p - (p^2 - 1) > 1$ has no prime factor $\leq p$, and $q_2 \leq N - p_{k+1} < p_{k+1}^2$, so q_2 is prime. □

*Repository: `github.com/shoblaster/Goldbach-Researchs`. AI-assisted drafting; all numbers are reproducible from the repository.

Hence Conjecture 1 implies Conjecture 2 for $k \geq 3$; the range $12 \leq N \leq 29$ ($k = 2$) is a finite check. The source's case split is written with thresholds in $n = N/2$ that behave like thresholds in N; this is a harmless boundary slip (repository note `PROOF_FRONTIER_022` §3; read from the arXiv text via web extraction, not a local PDF).

3 Exhaustive certification of $h_2(k)$ for $k \leq 11$

Method. For fixed (k, L) we search for a choice of two classes per prime covering $[1, L]$ by depth-first search: branch on the lowest uncovered position q and on which prime covers it with residue $q \bmod p$, pruning when an exact upper bound on the coverage still obtainable falls short. Certification of $h_2(k) = H$ means: $L = \omega_2(k)$ is feasible (witness re-verified directly against the definition) and $L = \omega_2(k) + 1$ is infeasible (search exhausted). No MILP/SAT solver is used. A parallel variant splitting at depth 2 gives identical infeasible node counts for $k = 5, \ldots, 10$; the $k = 11$ run used it (about 30 min on two threads).

k	p_k	$h_2(k)$	ω_2	$p_k^2 - p_k$	ratio	status
3	5	18	2	20	0.900	certified
4	7	30	4	42	0.714	certified
5	11	66	10	110	0.600	certified
6	13	150	24	156	**0.962**	certified
7	17	192	31	272	0.706	certified
8	19	258	42	342	0.754	certified
9	23	366	60	506	0.723	certified
10	29	450	74	812	0.554	certified
11	31	570	94	930	0.613	certified†
12–21	37–73	708–2622			≤ 0.60	transcribed

Table 1: Values from [1] Table 1; certified rows re-derived by exhaustive search († $k = 11$: one implementation, repository EXP032; EXP030; $h_2(10)$ also by an independent implementation, EXP029, same algorithm family). Full data: `data/h2_table.csv`.

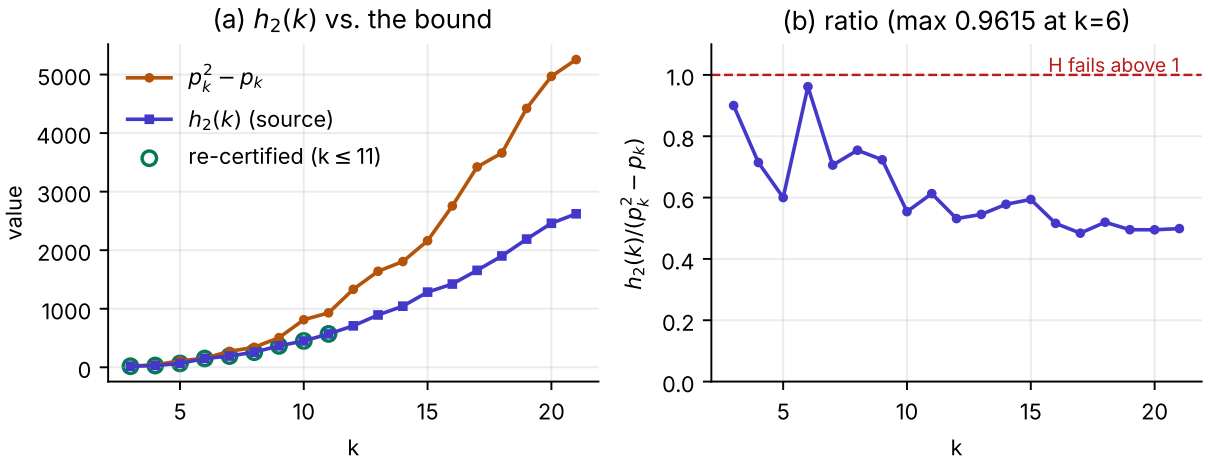


Figure 1: (a) $h_2(k)$ against $p_k^2 - p_k$; circles mark certified rows. (b) The ratio never exceeds 0.9615 ($k = 6$), so Conjecture 1 holds on all 19 tabulated rows, but the margin at $k = 6$ is only 6.

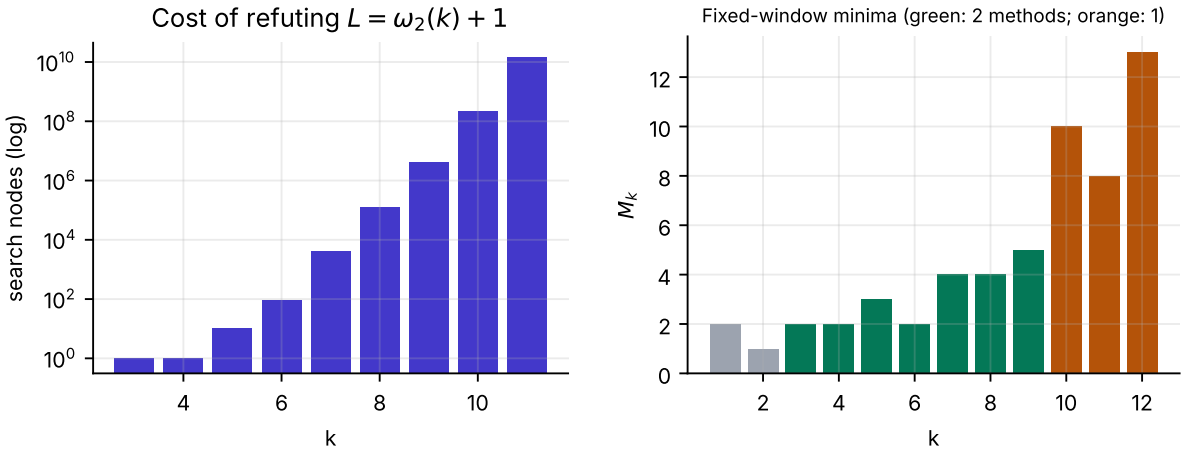


Figure 2: Left: nodes to refute $L = \omega_2(k) + 1$ for $k = 3..10$ (exact counts in `data/infeasibility_nodes.csv`; 2.1×10^8 at $k = 10$, 1.4×10^{10} at $k = 11$), growth $\approx \times 30$–65 per step. Right: exact fixed-window minima M_k (survivors in $[1, p_k^2]$ with one class mod 2 and two mod each odd prime); $k \le 9$ confirmed by two independent methods (a MILP and a solver-free branch and bound), $M_{10} = 10$, $M_{11} = 8$, $M_{12} = 13$ by the branch and bound alone.

Not obtained. $k \ge 12$: extrapolating the growth, $k = 12$ needs about 10^{12} nodes (roughly a day on two threads); lower-side runs for $k = 12, 13$ timed out at 240 s (EXP031).

Finite check of Conjecture 2. All 9,999,995 even $N \in [12, 2 \cdot 10^7]$ pass; an independent trial-division check for $N \le 10^5$ agrees (EXP023). This is evidence, not proof.

4 A dimension-2 sieve barrier (heuristic)

The covering window has length $\approx p_k^2$ and sifts by all primes $\le p_k \approx x^{1/2}$ with two classes removed per prime, i.e. sieve dimension $\kappa = 2$. A lower-bound sieve of dimension κ gives a positive count only when the sifting level satisfies $z \le x^{1/\beta_\kappa}$ with β_2 the sifting limit; Franze reports $\beta_2 = 4.266$ (Diamond–Halberstam–Richert) and 4.516 ($\Lambda^2\Lambda^-$), and cites $\beta_2 < 4.45$ (Blight) [5]. Thus the reachable range is $z \le x^{0.234}, x^{0.225}, x^{0.221}$, short of the $x^{1/2}$ required (Fig. 3). This is an argument by comparison with source-reported constants—not a theorem—and the parity problem [6] is the underlying obstruction. A bound of the Iwaniec type for the ordinary Jacobsthal function [4] would settle the order of h_2 only if its constant is explicit and small; a paired analogue is not known to us.

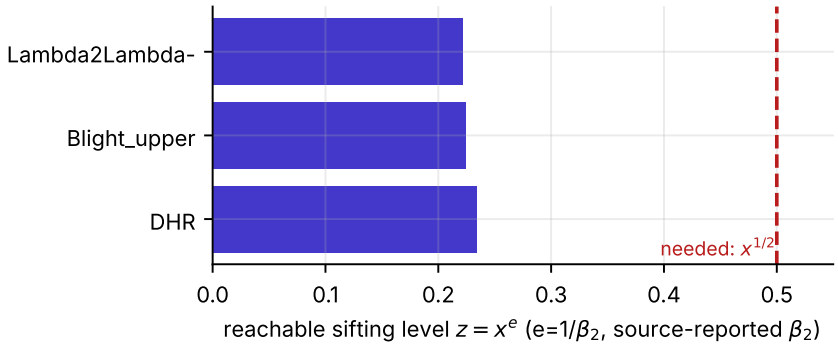


Figure 3: Reachable sifting exponent $1/\beta_2$ vs. the $x^{1/2}$ requirement. The β_2 values are *source-reported and not re-derived*.

5 Reproducibility

Every figure is produced by `11_PAPER/make_figures.py` from the CSVs in `11_PAPER/data/`; `verify_data.py` recomputes p_k, checks $h_2(k) = 6\omega_2 + 6$, $h_2 < p_k^2 - p_k$, and re-runs the search for $k \leq 8$, matching the node counts (about one minute on one core). Larger rows use `03_CODE/h2_cover_search(_par).cpp` (repository branch history; raw output in `06_EXPERIMENTS/EXP030*`). The parallel $k = 10$ run takes about 21 s.

6 Limitations and requests for review

(1) Conjecture 1 is unproved for all k; $k \geq 12$ is transcribed. (2) The certification trusts the covering-model equivalence from [3]; both implementations share one algorithm, so agreement is an implementation, not a method, cross-check; the $k = 11$ result has no independent re-run. (3) M_{10}, M_{11}, M_{12} rest on one implementation with no second method. (4) The β_2 values and the sieve comparison need checking against the primary books. (5) Novelty is not established. We would value a pointer to prior work on exact $h_2(k)$ or on paired-progression covering bounds, and on whether a structural route (symmetry, density of the extremal covers) could beat brute force beyond $k = 11$.

References

[1] M. Ziller, J. F. Morack, *A short note on the computation of the generalised Jacobsthal function for paired progressions*, arXiv:1706.03668 (2017).

[2] M. Ziller, J. F. Morack, *Divisibility in paired progressions, Goldbach's conjecture, and the infinitude of prime pairs*, arXiv:1706.00317 (2017).

[3] M. Ziller, J. F. Morack, ancillary file *On the computation of the generalised Jacobsthal function for paired progressions*, arXiv:1706.03668v1.

[4] L. Costello, D. Watts, *A short note on Jacobsthal's function*, arXiv:1306.1064 (2013).

[5] C. Franze, *Sifting limits for the* $\Lambda^2\Lambda^-$ *sieve*, arXiv:1012.3809 (2010).

[6] K. Ford, *On Bombieri's asymptotic sieve*, arXiv:math/0401215.